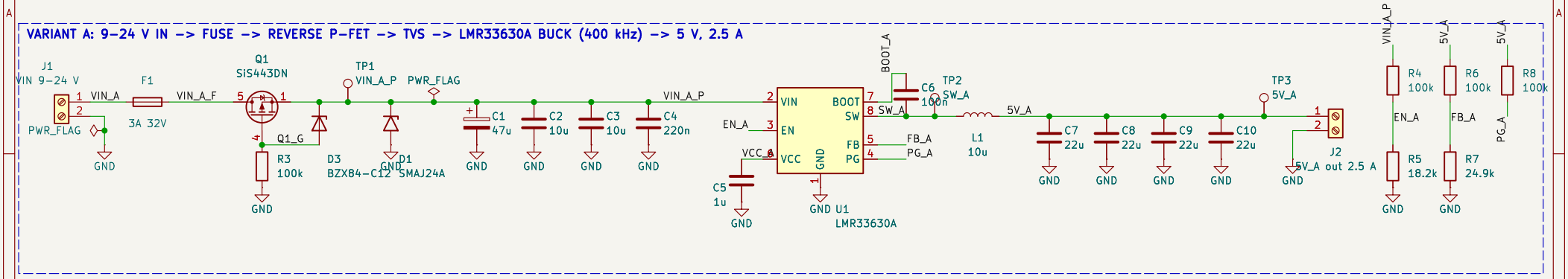
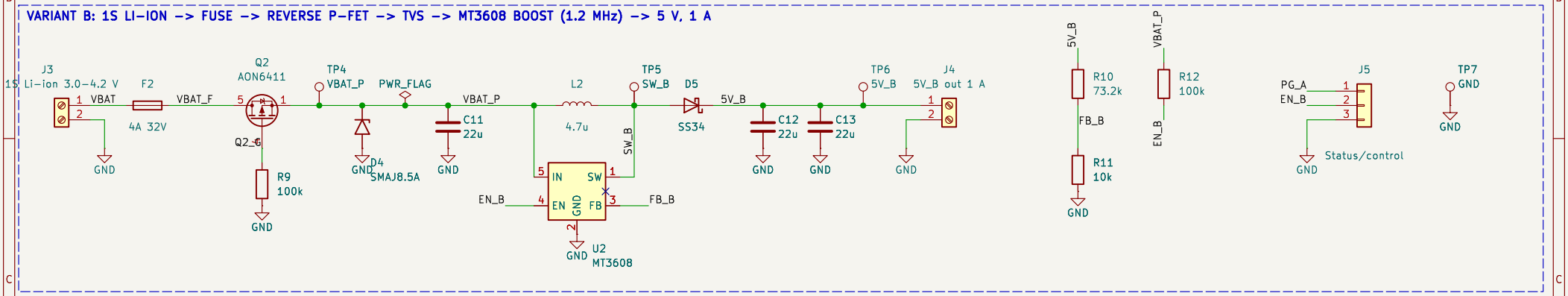
WARNING: Reference design. Test first with a current-limited supply and protection. Not for unattended use.

Design only, not fabricated, not bench-tested. Variant B: the Li-ion pack MUST have its own protection (over-discharge, short circuit).





KEY VALUES (README has the derivations and sources)

A: Vout = 1.0 V x (1 + 100k/24.9k) = 5.02 V (4.86–5.17 V worst case). L1 10 uH: ripple <= 1.46 A, peak <= 3.23 A at 2.5 A (HS limit >= 3.85 A).

A: EN divider 100k/18.2k starts at 7.8–8.2 V. Q1 |VGS| = VIN up to ~11.4 V, then D3 clamps (VGS max 20 V). F1 3 A covers 1.63 A at 9 V with 70 C derating.

A: SMAJ24A clamps at 38.9 V at full pulse, above the LMR33630 VIN abs max of 38 V. Input 9–24 V only.

B: Vout = 0.6 V x (1 + 73.2k/10k) = 4.99 V. Battery 1.96 A at 3.0 V / 1 A out (eta 0.85 assumed). MT3608 limit 4 A typ, no minimum given.

B: EN_B low stops switching but does NOT disconnect: VBAT - VF still reaches 5V_B. MT3608 UVLO is ~2 V: pack protection required.

B: SMAJ8.5A chosen over SMAJ5.0A for leakage (20 uA vs 800 uA max at VR). F2 4 A; a Li-ion short can exceed its 50 A interrupting rating.

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Abdulazeez Bello

Sheet: /
File: bus-5v.kicad_sch

Title: 5 V bus: 9–24 V buck and 1S Li-ion boost variants

Size: A4	Date: 2026-10-18	**Rev: A**
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